Do Now · retrieval

Data Structures · Classes and Objects in C#

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What is a method parameter?

Q2

What is a breakpoint?

Q3

How does a queue differ from stack?

Q4

Evaluate (A AND NOT B) OR C when A=1, B=0 and C=0. Show the intermediate Boolean values.

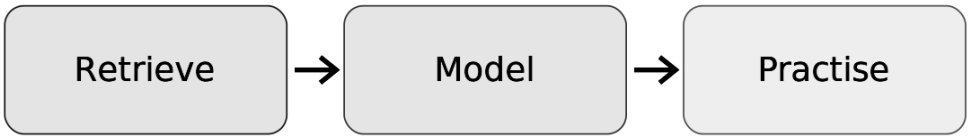
New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Class ≠ instance — A class defines the fields and behaviour available to each object. new Student() constructs an instance; a separate new creates another independent object.

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

class — Blueprint for objects

object — An instance of a class

field — Data associated with an object

method — Behaviour in a class

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

One class, two independent objects

Follow the worked method

1. Identify class, objects and members — Badge is the class (blueprint). amber and blue are references to two objects . Each has its own Points field and can run the Add method.

2. Trace the state change — new Badge() creates each object with Points = 0 . amber.Add(2) changes amber’s field from 0 to 2; blue remains at 0 because it is a different instance.

3. Predict, then check the output — The output is 2,0 . A class definition does not create an object. A method call on one object does not automatically change the fields of another.

Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

Trace both instances
Start with separate fields: first = 0, second = 0. Track which object receives each method call before revealing the result.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Apply independently

Write C# to define a class Book with public string Title. [3]

Question 2

Apply independently

Instantiate a Book object named novel. [2]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

Apply independently

Explain why two objects of the same class may hold different field values. [2]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

True or false: defining a class automatically creates an object.

Use key vocabulary from memory

1. Explain class in your own words.

2. Explain object in your own words.

3. Explain field in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.